method of condensing the information of the dataset into smaller graphs and relational columns.

This also helps shed light on the importance of the columns. Contrary to popular belief, it's not always preferable to include all the inputs in the model dataset. The reason to this being that not all inputs have equal weights and contributions to the output. So how do we check which columns to remove and keep? Principal Component Diagrams can be used to normalize and reduce the inputs into a collection of equations with linear inputs. But a simpler method would be to calculate ANOVA (Analysis of Variance) and ANCOVA (Analysis of Covariance) between the input and output. Then, perform a hypothesis test to check for significance at a particular interval level.

Take care and remember that certain inputs may need to be included in the model due to the nature of the experiment demanding its presence to check for effects and contributions. In this, disqualifying the columns is out of the question. Remember the adage, garbage in is garbage out.

OTHER TIPS FOR DATA PREPARATION

1. Fairly simple – start with a random sample of your data that has to be tested for exploratory analysis, instead of the entire dataset. Not only does this reduce computational time and energy, it also keeps any errors and redundancies in check. Imagine performing the same recurring actions on a very large data set and getting results ages after the test was performed, only to find a bug in testing. When in doubt, start small and compact.
2. Use simple column names to allow users and other stakeholders to understand what the columns indicate. If the columns are split over various datasets in multiple sources or servers, include a small index that explains what each column represents. A small issue in interpreting datatypes can yield unexpected consequences in the future.
3. Graphs speak better than numbers. It's amazing how data visualization softwares have come to exponentially reduce large datasets to drawings. Don't go for the complicated graphs that look better for results and employ simpler tools like control charts, bar charts, scatter plots, tree diagrams and, pie charts and histograms.
4. When profiling the data, have a "ballpark range" of values that relate to the possible values that the columnar records can take. Knowing this beforehand will cut down on the time taken to look for null values and outliers.